

Hands-on Lab: Joins in MySQL using phpMyAdmin



Estimated time needed: 20 minutes

In this lab, you will learn how to create tables and load data in the MySQL database service using the phpMyAdmin graphical user interface (GUI) tool.

Software Used in this Lab

In this lab, you will use [MySQL](#). MySQL is a Relational Database Management System (RDBMS) designed to efficiently store, manipulate, and retrieve data.



To complete this lab you will utilize MySQL relational database service available as part of IBM Skills Network Labs (SN Labs) Cloud IDE. SN Labs is a virtual lab environment used in this course.

Objectives

After completing this lab, you will be able to:

1. Determine the correct type of join to use for a given problem.
2. Write and execute joins to query data from multiple tables.

Database Used in this Lab

mysql_learners database has been used in this lab.

Here you will be creating and inserting data into the below mentioned 3 tables

- 1.chicago_public_schools
- 2.chicago_socioeconomic_data
- 3.chicago_crime

Here you will be using 3 dump files for this purpose.

[chicago_public_schools](#)

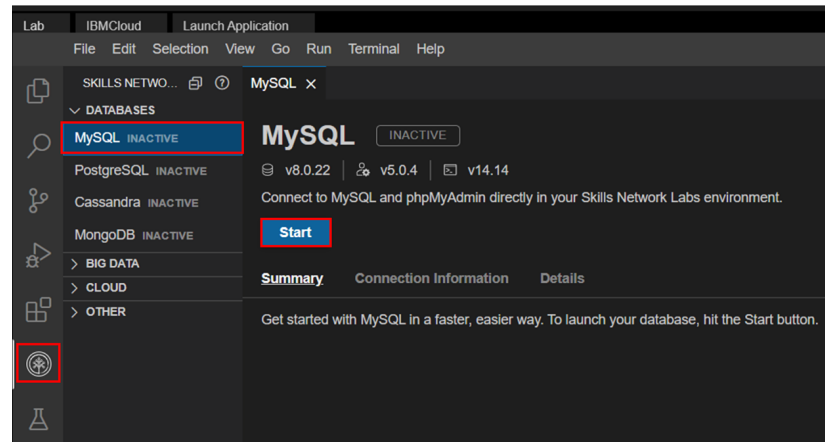
[chicago_crime](#)

[chicago_socioeconomic_data](#)

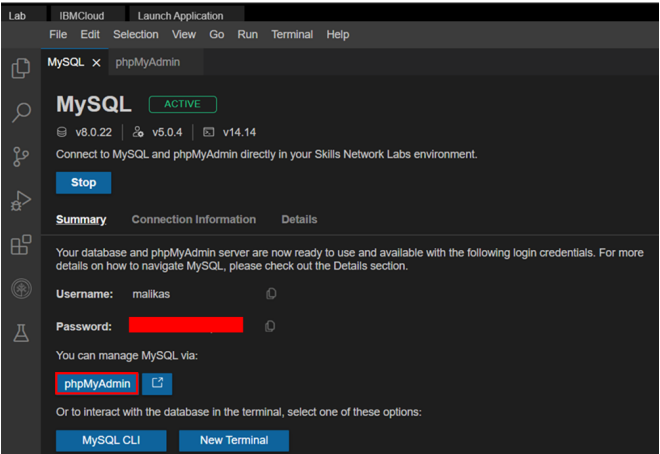
Task A: Create a database

1. Click on **Skills Network Toolbox**. In **Database** section, click **MySQL**.

To start the MySQL click **Start**.



2. Once **MySQL** has started, click on **phpMyAdmin** button to open **phpMyAdmin** in the same window.



3. You will see the phpMyAdmin GUI tool.

phpMyAdmin

Recent Favorites

New

- information_schema
- mysql
- performance_schema
- sakila
- sys

Server: mysql:3306

Databases SQL Status User accounts Export Import Settings Binary log Replication VariablesCharsets Engines

General settings

Server connection collation: utf8mb4_unicode_ci

[More settings](#)

Appearance settings

Language: English

Theme: pmahomme

Database server

- Server: mysql via TCP/IP
- Server type: MySQL
- Server connection: SSL is not being used
- Server version: 8.0.22 - MySQL Community Server - GPL
- Protocol version: 10
- User: root@172.18.0.2
- Server charset: UTF-8 Unicode (utf8mb4)

Web server

- Apache/2.4.38 (Debian)
- Database client version: libmysql - mysqlnd 7.4.15
- PHP extension: mysqli curl mbstring
- PHP version: 7.4.15

phpMyAdmin

- Version information: 5.0.4, latest stable version: 5.1.0
- [Documentation](#)
- [Official Homepage](#)
- [Contribute](#)
- [Get support](#)
- [List of changes](#)
- [License](#)

4. In the tree-view, click **New** to create a new empty database. Then enter **MySQL_Learners** as the name of the database and click **Create**.
The encoding will be left as **utf8mb4_0900_ai_ci**. UTF-8 is the most commonly used character encoding for content or data.
Proceed to Task B.

DatabasesSQLStatusUser accountsExportImportSettingsBinary logReplicationVariablesCharsetsMore

Databases

Create database

Mysql_learnersutf8mb4_0900_ai_ciCreate

Database	Collation	Master replication	Action
☐ information_schema	utf8_general_ci	✔ Replicated	Check privileges
☐ mysql	utf8mb4_0900_ai_ci	✔ Replicated	Check privileges
☐ performance_schema	utf8mb4_0900_ai_ci	✔ Replicated	Check privileges
☐ sys	utf8mb4_0900_ai_ci	✔ Replicated	Check privileges

Total: 4

⬆

☐ Check all

With selected:

[Drop](#)

⚠ Note: Enabling the database statistics here might cause heavy traffic between the web server and the MySQL server.

[Enable statistics](#)

Load the dump files one by one into the database **Mysql_learners** by clicking the **Import** tab and choose the file. Click on **Go** button.

Importing into the database "Mysql_learners"

File to import:

File may be compressed (gzip, bzip2, zip) or uncompressed.
A compressed file's name must end in **.[format].[compression]**. Example: **.sql.zip**

Browse your computer:

Choose File

chicago_pu...schools.sql (Max: 2,048KiB)

You may also drag and drop a file on any page.

Character set of the file:

utf-8

Partial import:

☒ Allow the interruption of an import in case the script detects it is close to the PHP timeout limit. *(This might be a good way to import large files, however it can break transactions.)*

Skip this number of queries (for SQL) starting from the first one:

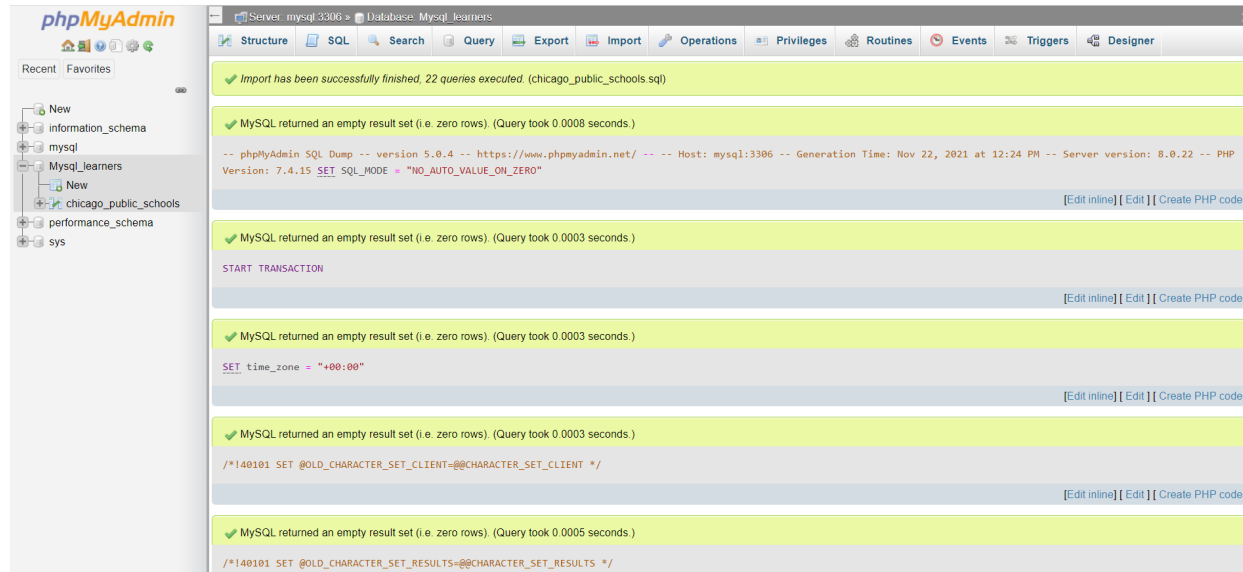
0

Other options:

☒ Enable foreign key checks

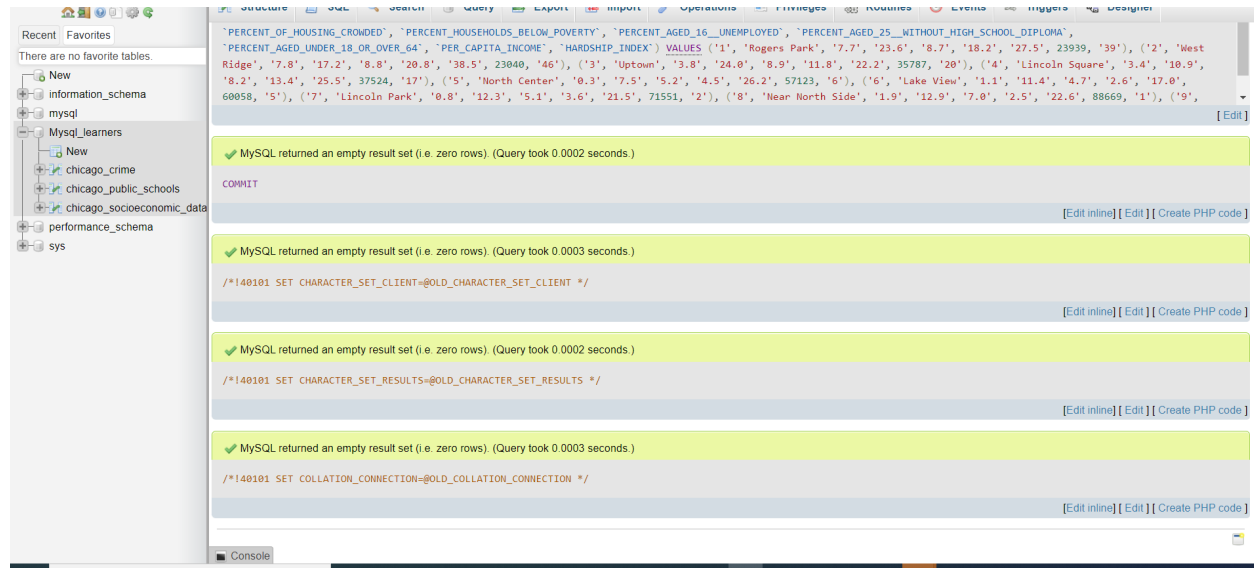
Format:

Console



The tables are created and the data is loaded successfully. Repeat the same operation with the other 2 dump files to create and load the tables.

You will see a screen as below



Exercise

Problem 1

List the case number, type of crime and community area for all crimes in community area number 18.

- Hint 1
Use tables `chicago_public_schools` and `chicago_socioeconomic_data`.
- Hint 2
Use an inner join.
- Hint 3
The column `PRIMARY_TYPE` contains the crime type.

Problem 2

List all crimes that took place at a school. Include case number, crime type and community name.

- Hint 1
Use tables `chicago_crime` and `chicago_socioeconomic_data`.
- Hint 2
Use a left join or a right join.
- Hint 3
The column `LOCATION_DESCRIPTION` will help you find the crime location.

Problem 3

For the communities of Oakland, Armour Square, Edgewater and CHICAGO list the associated `community_area_numbers` and the `case_numbers`.

- Hint 1
Use tables `chicago_crime` and `chicago_socioeconomic_data`.
- Hint 2
Use the union operator with `LEFT` and `RIGHT` outer joins.
- Hint 3
Use `COMMUNITY_AREA_NUMBER` from `chicago_socioeconomic_data`.

Congratulations! You have completed this lab, and you are ready for the next topic.

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